How Much of a Time Series Survives Its Image? A Cross-Domain Comparative Analysis of the Reconstruction Capacity of Image-Based Representations

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Abstract

Image-based representations of time series are widely used as classifier inputs and, increasingly, as the space in which signals are generated or restored, which requires mapping an image back to a series. We present a comparative analysis of how much of a time series each representation retains and can return. We study the most widely used representations, namely the recurrence plot (RP) in its binary, unsigned and signed forms, the Gramian angular summation and difference fields and the Markov transition field, together with the recently proposed extended intertemporal return plot (XIRP), whose diagonal stores the series itself. Each is decoded from its stored 8-bit image alone, with the published inverse where one exists and an operational decoder where none does, and decoding the floating-point matrix separates the loss of the representation from that of storage. On seven datasets from five domains and windows of 64 to 256 samples we measure temporal and spectral fidelity, computational cost, the conditions under which reconstruction is exact, and whether classifiers trained on original windows keep their accuracy on the

reconstructions. [pending: key findings]

Keywords: Time-series imaging, Reconstruction, Recurrence plots, Gramian angular fields, Markov transition fields, Invertibility, Comparative analysis

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1. Introduction

Turning a time series into an image has become a common way of giving image-domain models access to temporal data. Recurrence plots (RP), Gramian angular fields (GAF) and Markov transition fields (MTF) place a relation between every pair of time instants in a matrix, and convolutional and other image models then learn from that matrix [1, 2, 5, 6]. The same representations are now also used in the opposite direction: a generative or restoration model works on images, and its output has to be turned back into a series [3, 4].

That second use depends on a property the first never needed. A classifier only has to separate classes, and it can do so from an image that has discarded amplitude, phase or temporal order. A model that produces images is useful only if those images can be decoded into the signal they stand for. Whether, and how well, each representation allows this has not been compared: reconstruction is usually studied for one representation at a time, on one kind of signal, and is often judged only through the accuracy of a downstream classifier.

This paper is a comparative analysis of the reconstruction capacity of image-based time-series representations. We take the most widely used ones, the recurrence plot, the Gramian angular summation and difference fields (GASF, GADF) and the Markov transition field, and add the extended

intertemporal return plot (XIRP) of Hellermann and Lessmann [4], a recent representation proposed for image-based generation that stores the series itself on its diagonal and is therefore recoverable by construction. For the recurrence plot we separate the encoding from the decoder: we compare the binary RP of the classical literature, the continuous unsigned RP and the signed RP of Lu and Tong [11], decoded both with the classical reconstruction of Hirata et al. [12] and with a geometric decoder.

The setting is controlled. Every image is the deterministic output of a known encoder, stored with 8 bits per pixel, and every decoder works from the stored image alone. Where a representation has a published inverse we use it; where it has none that can be applied to a stored image, as for the signed RP and GADF, we provide one, since otherwise the comparison could not be made. Decoding the floating-point matrix as well as the 8-bit image separates what the representation loses from what storage loses.

The analysis answers five questions:

- *RQ1.* How much of the signal can be recovered from each representation, in time and in frequency, and how much of the loss is due to 8-bit storage?
- *RQ2.* For recurrence plots, what do the encoding and the decoder each contribute?
- *RQ3.* Under which conditions is reconstruction exact, and how does it vary with the domain, the sampling rate and the window length?
- *RQ4.* Do reconstructions preserve what classifiers trained on the original signals need?

- *RQ5.* What does each reconstruction cost in time and memory?

The contributions are:

- A comparative analysis of the reconstruction capacity of five image representations, eight encoding–decoder combinations in total, on seven public datasets from five domains (human activity, environmental audio, electrocardiography, geomagnetism and seismology) and three window lengths, under one protocol of fidelity, cost and classification utility.
- A separation of the two ingredients of RP reconstruction, encoding and decoder, which shows what the sign of the signed RP adds and what the decoder adds.
- A statement of what each image determines and of the conditions under which it is exactly, approximately or only partially invertible, with the conditioning of the recurrence-plot states identified as the quantity that governs reconstruction from the signed RP after storage.
- Operational, metadata-free decoders for the two representations that lacked one, the signed RP and GADF, and the code and derived data needed to reproduce every result.

2. Related work

2.1. Image representations of time series

Recurrence plots were introduced to visualise when a dynamical system revisits a region of its state space [5] and have since been developed into a

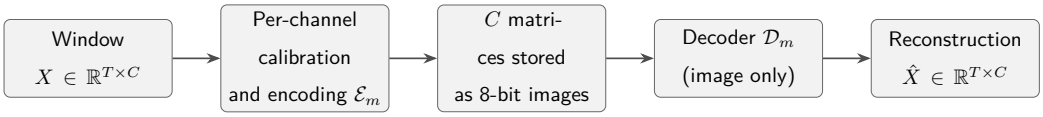


Figure 1: Controlled reconstruction. Each channel is calibrated, encoded and stored as an 8-bit image, then decoded from the image alone; the shared per-channel calibration returns it to physical units. Decoding the floating-point operator instead of the image isolates the algorithmic error from the storage error.

family of quantitative tools [6]. In their conventional form they are binary: a pixel marks whether two embedded states lie within a threshold of each other. Continuous variants keep the distance itself, and the signed variant of Lu and Tong also keeps whether the displacement between two states follows a reference direction, which they used for activity recognition from a single accelerometer [11]. Wang and Oates proposed the Gramian angular fields, which map each sample to an angle and store the cosine of their sum or the sine of their difference, and the Markov transition field, which stores transition probabilities between quantile states, as image inputs to convolutional networks [1, 2]. These representations have been applied to human activity recognition [7], anomaly detection [8] and imputation [9]. Hellermann and Lessmann recently introduced XIRP for time-series generation with image-based generative adversarial networks; it stores log-returns between instants off the diagonal and the series on the diagonal, so that generated images can be inverted [4].

2.2. Recovering a series from its image

Inversion has been studied separately for each representation. For binary recurrence plots, Hirata et al. build a graph from the recurrence neighbour-

hoods, approximate the distances between states by shortest paths and recover the state geometry by multidimensional scaling [12][cite: Hirata et al. 2016, Sci. Rep. 6:34982]. For unthresholded recurrence plots of delay embeddings, Sipers et al. characterise when the signal is identifiable up to its mean and sign and reconstruct it by optimisation [13, 14]. For GASF, Wang and Oates read the series from the diagonal, and restore damaged images with a denoising autoencoder before doing so [1]; Tao et al. follow a similar route for RP and GAF images [9]. For GADF no decoder that resolves its branch ambiguity and its lost global offset has been specified, and for MTF even approximate recovery has been reported to be difficult [1]. XIRP is inverted from its diagonal or by averaging the proposals of all its columns [4].

2.3. Gap

Each of these works evaluates one representation, usually on one kind of signal, and several evaluate reconstruction only through classification. No study has compared the reconstruction capacity of the representations under a common protocol, separated what a representation loses from what image storage loses, or distinguished, for recurrence plots, the contribution of the encoding from that of the decoder.

3. Mathematical formulation

3.1. Encoding, storage and decoding

Let $x = (x_1, \ldots, x_T)$ be one channel of a window. Every method m applies a calibration to a method-specific normalised domain, an encoder $\mathcal{E}_m$ producing a matrix $M^{(m)}$, a fixed 8-bit storage map $\mathcal{Q}_m$ and a decoder $\mathcal{D}_m$:

$$\hat{x} = \mathcal{N}^{-1}\Big(\mathcal{D}_m(\mathcal{Q}_m(\mathcal{E}_m(\mathcal{N}(x)))); \, x_{\max}, x_{\min}\Big).$$

The calibration $(x_{\max}, x_{\min})$ is shared by all methods and is the only per-channel information used to return to physical units. We call a representation *exactly invertible* when $\mathcal{D}_m \circ \mathcal{E}_m$ is the identity on its normalised domain, so that all error in $\hat{x}$ is due to $\mathcal{Q}_m$; *approximately invertible* when $\mathcal{D}_m \circ \mathcal{E}_m$ is not the identity but recovers the series up to a bounded algorithmic error; and *lossy* when $\mathcal{E}_m$ is many-to-one.

3.2. Recurrence plots

With the adjacent states $s_i = (x_i, x_{i+1})$, $i = 1, \ldots, T-1$, and $v = (1,1)/\sqrt{2}$, the continuous signed RP [11] is

$$R_{ij} = \sigma_{ij}\|s_i - s_j\|_2, \qquad \sigma_{ij} = \begin{cases} +1, & \|s_i - s_j\| = 0 \text{ or } \hat{u}_{ij} \cdot v \geq \cos(\pi/4), \\ -1, & \text{otherwise}, \end{cases} \tag{1}$$

with $\hat{u}_{ij} = (s_i - s_j)/\|s_i - s_j\|$. The unsigned RP is $|R|$, and the conventional binary RP is $B_{ij} = \mathbb{1}[\|s_i - s_j\| < \varepsilon]$ [5, 6]. A pair carries directional information only when its displacement lies within $\pi/4$ of $\pm v$: then $\sigma_{ij} = -\sigma_{ji}$, while displacements near the anti-diagonal receive -1 in both orders.

Proposition 1 (Signed RP, floating point)**.** *Let $S \in \mathbb{R}^{(T-1)\times 2}$ be the matrix of states and assume that the centred states span $\mathbb{R}^2$, i.e. $\sigma_2(S - \bar{S}) > 0$. Then classical MDS applied to $|R|$ recovers S up to a translation and an orthogonal transformation $Q \in O(2)$. If, in addition, the adjacent-state overlap $b_i = a_{i+1}$ is satisfied by a unique orientation up to the sign selection of Section 4.2, the decoder returns x exactly on the normalised domain.*

Proof sketch. The first part is the classical result that an exact Euclidean distance matrix of points with affine rank two is reproduced up to isometry by

the two leading eigenvectors of the doubly centred squared-distance matrix [15]. For the second, the overlap residual is linear in $(\cos\theta, \sin\theta)$ for each reflection parity (Section 4.2), so the exact orientation attains zero residual. [pending: complete proof and the uniqueness condition, issue #121]

Remark 1 (8-bit storage)**.** *Storage perturbs $|R|$ by at most half a quantisation step relative to its range. The error of the recovered two-dimensional subspace grows as the gap between the second and third eigenvalues of the centred Gram matrix shrinks, so the reconstruction degrades as σ_2/σ_1 of the centred states tends to zero, i.e. as the states become collinear[cite: eigenvector perturbation bound, e.g. Davis–Kahan]. Empirically, no series with $\sigma_2/\sigma_1 \geq 0.1$ exceeds a normalised error of* 0.01 *(Section 6.3).*

3.3. Gramian angular fields

After scaling to the method-specific interval and $\phi_i = \arccos \tilde{x}_i$, the summation and difference fields are $G^{\mathrm{S}}_{ij} = \cos(\phi_i + \phi_j)$ and $G^{\mathrm{D}}_{ij} = \sin(\phi_i - \phi_j)$ [1]. On $[0, 1]$, $G^{\mathrm{S}}_{ii} = \cos(2\phi_i)$ determines $\tilde{x}_i$ uniquely, so GASF is exactly invertible. GADF has $G^{\mathrm{D}}_{ii} = 0$; on the symmetric interval $[-\alpha, \alpha]$ the inverse sine returns $\phi_i - \phi_j$ without branch ambiguity if and only if $\alpha < 1/\sqrt{2}$, and the remaining global offset is fixed by an image-derived boundary anchor (Section 4.4). Under these hypotheses GADF is exactly invertible.

3.4. Markov transition fields and XIRP

MTF assigns every sample to one of Q quantile states and stores the transition probability between the states occupied at two positions [1]. Samples in the same state are indistinguishable, so the operator is many-to-one and only a representative trajectory can be recovered. XIRP [4] requires a strictly

Table 1: Information determined by each image and the resulting class of invertibility. Every decoder uses the image alone on the normalised domain; returning to physical units uses the shared calibration $(x_{\max}, x_{\min})$.

Representation	Image determines	Invertibility
Signed RP	State geometry and directional sign	Exact if $\sigma_2/\sigma_1 > 0$ (Prop. 1)
Unsigned RP	State geometry only	Exact up to reflection
Binary RP	Recurrence neighbourhoods	Approximate (graph geodesics)
GASF	Angles on one branch	Exact on $[0, 1]$
GADF	Angular differences	Exact if $\alpha < 1/\sqrt{2}$
MTF	Quantile states and transitions	Lossy
XIRP	Series on the diagonal	Exact

positive series and stores $R_{ii} = x_i$ and $R_{ij} = \log(x_i/x_j)$; it is exactly invertible from its diagonal, and inversion by mean averages the T column proposals $x_j \exp(R_{ij})$.

3.5. What each image determines

Table 1 summarises, for each representation, what the image determines, what the decoder needs, and its class of invertibility.

4. Reconstruction methods

Table 2 lists the decoder used for each representation in the comparison. Published inverses are used as described by their authors, with only the

Table 2: Encoding and decoder of every compared combination.

Encoding	Decoder	Origin of the decoder
Binary RP	Graph geodesics and MDS	Hirata et al. [12]
Signed RP	Hirata, sign used for the reflection	Hirata et al., adapted
Unsigned RP	Two-dimensional MDS and overlap	This work
Signed RP	Two-dimensional MDS, overlap and sign	This work
GASF	Diagonal inverse	Wang and Oates [1]
GADF	Restricted branch and boundary anchor	This work
MTF	Fiedler-ordered representative	This work (lossy)
XIRP	Inversion by mean	Hellermann and Lessmann [4]

adaptations needed to decode a stored 8-bit image; the signed-RP and GADF decoders are ours, because neither representation had a decoder that could be applied to a stored image.

4.1. Storage protocol

Each matrix is written to $[0, 255]$ by a fixed or per-window affine map. GASF and GADF are read back with the fixed operator range $[-1, 1]$, MTF with $[0, 1]$, and XIRP with the fixed ranges of its diagonal and off-diagonal parts after the series is mapped to $[1, 2]$. The signed RP keeps a per-window map; its zero-valued diagonal locates $R = 0$, so subtracting the median

diagonal pixel recovers the signed matrix up to a positive scale that MDS and the final range map absorb. No decoder uses per-window metadata.

4.2. Signed-RP decoder

1. Recover $R^{\text{obs}} = \text{PNG} - \text{median}_i \, \text{PNG}_{ii}$ and the symmetric distances $D_{ij} = \frac{1}{2}(|R^{\text{obs}}_{ij}| + |R^{\text{obs}}_{ji}|)$.
2. Classical MDS [15]: with $J = I - \mathbf{1}\mathbf{1}^\top/(T-1)$, the two leading eigenpairs of $-\frac{1}{2}JD^{\circ 2}J$ give $Y = V_2\Lambda_2^{1/2}$.
3. Orientation in closed form. Every $Q \in O(2)$ is $\text{rot}(\theta)\,\text{diag}(1, \eta)$ with $\eta \in \{\pm 1\}$. Writing $Z = YQ$ and $(a_i, b_i) = (Z_{i1}, Z_{i2})$, the overlap $b_i = a_{i+1}$ must hold up to a common translation δ. For fixed η the residual $b_i - a_{i+1} = A_i \cos\theta + B_i \sin\theta$ with $A_i = \eta Y_{i2} - Y_{i+1,1}$ and $B_i = -\eta Y_{i1} - Y_{i+1,2}$; profiling out δ centres A and B, and the objective becomes the Rayleigh quotient $u^\top H_\eta u$ with $H_\eta = \tilde{C}^\top \tilde{C}$, $\tilde{C} = [\tilde{A}\ \tilde{B}]$ and $u = (\cos\theta, \sin\theta)$. Its minimiser is the eigenvector of the smaller eigenvalue of the 2×2 matrix H_η, so the two parities give four exact candidates.
4. Sign selection. Among the four candidates, keep the one whose regenerated signed RP is closest to R^{obs} in Frobenius norm after both are normalised to unit norm. The orientation is chosen from the image alone; the original series is never consulted.
5. Samples: $\hat{x}_1 = a_1$, $\hat{x}_i = \frac{1}{2}(b_{i-1} + a_i)$ for $2 \le i \le T-1$, $\hat{x}_T = b_{T-1}$, and the minimum and maximum are mapped to the normalised range.

The orientation costs $O(T)$ and the decoder is dominated by the eigendecomposition, $O(T^3)$ in a dense implementation. On the unsigned RP the same decoder runs without step 4, so the reflection remains undetermined.

4.3. Classical reconstruction of Hirata et al.

As the reference for the conventional RP we implement the reconstruction of Hirata et al. [12][cite: Hirata et al. 2016]: from the binary RP, an edge joins $i \neq j$ when $B_{ij} = 1$, with local length $1 - |G_i \cap G_j|/|G_i \cup G_j|$, $G_i = \{k : B_{ik} = 1\}$; all-pairs shortest paths give a distance matrix, and classical MDS gives the states. The threshold ε is not prescribed by the method; we set it to a recurrence rate of 0.75. Turning the states into T samples is our post-processing: the leading MDS coordinate, the mean appended to reach T samples, and the range map. A binary RP carries no sign, so the reflection is left as returned. When Hirata's method is applied to the signed RP, its graph sees only the thresholded magnitude, so the sign is used, as in our decoder, only to choose between the series and its reflection.

4.4. GASF, GADF, MTF and XIRP decoders

GASF uses the diagonal inverse $\hat{\tilde{x}}_i = \cos(\frac{1}{2}\arccos G^{\mathrm{S}}_{ii})$ [1]. For GADF on $[-\alpha, \alpha]$ with $\alpha = 0.7$, the row means $\mu_i = \frac{1}{T}\sum_j G^{\mathrm{D}}_{ij}$ are strictly increasing in ϕ_i, so $e = \arg\max_i |\mu_i|$ identifies a boundary sample and the sign of μ_e fixes which boundary; the angles follow from $\hat{\phi}^{(p)}_j = \phi_e - \arcsin G^{\mathrm{D}}_{ep} - \arcsin G^{\mathrm{D}}_{pj}$, averaged over the five pivots with the smallest $|\mu_p|$. MTF ($Q = 4$) is decoded to a representative trajectory by clustering the rows into Q states, ordering them by the Fiedler vector of the symmetrised transition matrix [16] and assigning each state its bin midpoint. XIRP uses inversion by mean [4].

Table 3: Datasets. Windows of 128 samples available after segmentation; the group is the unit that never appears on both sides of a split.

Dataset	f_s (Hz)	Channels	Task	Windows	Group
WISDM	20	3	5 activities	8 172	subject (51)
PAMAP2	100	9	12 activities	15 085	subject (9)
ESC-50	8 000	1	50 sound classes	624 000	source recording
PTB-XL	100	12	5 classes, multilabel	21 000	patient
Swarm MAG	1	3	none	62 757	–
STEAD	100	3	earthquake vs noise	92 000	event / station
EMG gestures	100	8	6 gestures	880	subject (36)

5. Experimental protocol

5.1. Datasets

Table 3 lists the seven public datasets. They span human activity (WISDM [17], PAMAP2[cite: Reiss and Stricker 2012]), environmental audio (ESC-50[cite: Piczak 2015]), electrocardiography (PTB-XL[cite: Wagner et al. 2020]), geomagnetism (Swarm[cite: Swarm mission and data]), seismology (STEAD[cite: Mousavi et al. 2019]) and surface electromyography (EMG[cite: UCI EMG Data for Gestures, doi:10.24432/C5ZP5C]), with sampling rates from 1 Hz to 8 kHz and one to twelve channels.

5.2. Windows and splits

Every channel is encoded, stored and decoded independently, as its own single-channel 8-bit image with its own calibration, whatever the number of channels: one image for ESC-50, three for WISDM, Swarm and STEAD, and eight, nine and twelve for EMG, PAMAP2 and PTB-XL. Channels are neither

packed into colour planes nor combined, so results are comparable across datasets and cost grows linearly with the number of channels; representations that couple channels are outside the scope of this study. Each recording is cut into non-overlapping windows of $T \in \{64, 128, 256\}$ samples that never cross a label change or a time gap. For fidelity, 500 windows per dataset and length are drawn with a fixed seed. For classification, up to 6 000 windows per dataset and length are drawn, and three grouped folds keep every subject, patient, event, station or source recording on one side; folds are stratified by class for single-label data. Blocks with fewer than 200 windows (EMG at $T = 256$, where a gesture lasts about three seconds) are not reported. ESC-50 at 128 samples spans 16 ms of sound; audio is therefore also evaluated on the official ESC-10 subset at 2, 4 and 8 kHz with windows of 512 and 1 024 samples, silent windows removed and accuracy also scored per clip.

5.3. Fidelity, cost and utility

Fidelity is measured per channel series with range-normalised errors, writing $r = \max_i x_i - \min_i x_i$:

$$\mathrm{NMAE} = \frac{1}{rT} \sum_i |x_i - \hat{x}_i|, \qquad \mathrm{NRMSE} = \frac{1}{r} \sqrt{\tfrac{1}{T} \sum_i (x_i - \hat{x}_i)^2},$$

together with Pearson correlation, the range-normalised error of first differences, the L_1 distance between unit-mass Welch spectra [18], the dominant-frequency error and the energy error. Constant series are counted and excluded. Each decoder is run on the floating-point operator and on the 8-bit image, which separates algorithmic from storage error. Cost is the single-thread time to encode and decode and the peak traced memory, measured for T from 64 to 1 024.

Utility follows the train-on-original protocol: KNN, SVM, random forest, LSTM, GRU and a one-dimensional CNN are trained on original windows and evaluated on the original and on every reconstruction of the same held-out windows.

5.4. Statistical protocol

All comparisons are paired over the same series. Differences between decoders are summarised by medians with 95% bootstrap intervals that resample whole groups [21, 22], tested with the Wilcoxon signed-rank test [19] and Holm's correction within each family [25], and averaged across datasets with equal weight. Classification equivalence requires the 95% group-bootstrap interval of the accuracy difference to lie within ± 0.01 [23].

6. Results

6.1. Fidelity across representations and domains

[pending: Table 2: 8-bit NMAE of the eight decoders on seven datasets at $T = 64, 128, 256$, from results/_benchmark/summary.json and stats.json; Figure 4: reconstructions in each domain.]

6.2. Recurrence plots: decoder versus encoding

[pending: Figure 3 and Table 3, first block: Hirata and the proposed decoder on the unsigned and the signed RP, float and 8-bit, with the orientation oracle as a bound; from results/_stats/rp_encodings.json.]

6.3. When is the signed RP decoder exact?

[pending: Error against σ_2/σ_1 of the states, pooled over all datasets; from results/_stats/rp_conditions.json. No series with $\sigma_2/\sigma_1 \geq 0.1$ exceeds NMAE 0.01; Swarm is the collinear limit.]

6.4. Robustness and computational cost

[pending: Figure 5: robustness to image perturbation (#128) and time and memory against T (results/_stats/cost_scaling.json, #90).]

6.5. Preservation of classification utility

[pending: Table 3, second block: accuracy difference to the original for six classifiers on six labelled datasets and three lengths, with equivalence; ESC-10 per-clip accuracy; from results/_classification and results/_classification_esc10.]

7. Discussion

[pending: What the sign adds and what the decoder adds; conditioning and the collinear limit; sampling rate (#87); limitations: controlled inputs only, shared calibration, 8-bit storage, dataset coverage.]

8. Conclusions

[pending: After M10.]

CRediT authorship contribution statement

Adriano García-Giralda Milena: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Validation, Visualization, Writing – original draft, Writing – review & editing. **Francisco García Moreno:** Conceptualization, Methodology, Supervision, Writing – review & editing. **José Antonio Jiménez Madrid:** Methodology, Formal analysis, Writing – review & editing. **María Bermúdez Edo:** Supervision, Writing – review & editing.

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Data availability

All seven datasets are public and are cited at their official sources. The code, window and split manifests, configurations and derived results are available at `https://github.com/adrianoggm/TFG-Tratamiento-de-Series-Temporales` [pending: archived release with DOI, #137].

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

[pending: List every tool used for writing and for code, and what it was used for (#138).] All AI-assisted content was critically reviewed, verified and revised by the authors, who take full responsibility for the accuracy, originality and integrity of the manuscript.

References

[1] Zhiguang Wang and Tim Oates. "Imaging time-series to improve classification and imputation". In *Proceedings of the 24th International Joint Conference on Artificial Intelligence (IJCAI)*, pp. 3939–3945, 2015.

[2] Zhiguang Wang and Tim Oates. "Encoding time series as images for visual inspection and classification using tiled convolutional neural networks". In *Workshops at the Twenty-Ninth AAAI Conference on Artificial Intelligence*, 2015.

[3] Tongge Huang, Pranamesh Chakraborty, and Anuj Sharma. "Deep convolutional generative adversarial networks for traffic data imputation encoding time series as images". *International Journal of Transportation Science and Technology*, vol. 12, no. 1, pp. 1–18, 2023. doi: https://doi.org/10.1016/j.ijtst.2021.10.007.

[4] Justin Hellermann and Stefan Lessmann. "Leveraging image-based generative adversarial networks for time series generation". *International Journal of Forecasting*, vol. 42, no. 2, pp. 616–639, 2026. doi: https://doi.org/10.1016/j.ijforecast.2025.09.004.

[5] Jean-Pierre Eckmann, S. Oliffson Kamphorst, and David Ruelle. "Recurrence plots of dynamical systems". *Europhysics Letters*, vol. 4, no. 9, pp. 973–977, 1987. doi: https://doi.org/10.1209/0295-5075/4/9/004.

[6] Norbert Marwan, M. Carmen Romano, Marco Thiel, and Jürgen Kurths. "Recurrence plots for the analysis of complex systems". *Physics Reports*, vol. 438, no. 5–6, pp. 237–329, 2007. doi: https://doi.org/10.1016/j.physrep.2006.11.001.

[7] Zeeshan Ahmad and Naimul Khan. "Inertial sensor data to image encoding for human action recognition". *IEEE Sensors Journal*, vol. 21, no. 9, pp. 10978–10988, 2021. doi: https://doi.org/10.1109/JSEN.2021.3062261.

[8] Jidong Ma and Hairu Wang. "Anomaly detection in sensor data via encoding time series into images". *Journal of King Saud University – Computer and Information Sciences*, vol. 36, no. 10, article 102232, 2024. doi: https://doi.org/10.1016/j.jksuci.2024.102232.

[9] Huamin Tao, Qiuqun Deng, and Shanzhu Xiao. "Reconstruction of time series with missing value using 2D representation-based denoising autoencoder". *Journal of Systems Engineering and Electronics*, vol. 31, no. 6, pp. 1087–1096, 2020. doi: https://doi.org/10.23919/JSEE.2020.000081.

[10] Floris Takens. "Detecting strange attractors in turbulence". In *Dynamical Systems and Turbulence, Warwick 1980*, Lecture Notes in Mathematics, vol. 898, pp. 366–381, Springer, 1981. doi: https://doi.org/10.1007/BFb0091924.

[11] Jianjie Lu and Kai-Yu Tong. "Robust single accelerometer-based activity recognition using modified recurrence plot". *IEEE Sensors Journal*, vol. 19, no. 15, pp. 6317–6324, 2019. doi: https://doi.org/10.1109/JSEN.2019.2911204.

[12] Yoshito Hirata, Shunsuke Horai, and Kazuyuki Aihara. "Reproduction of distance matrices and original time series from recurrence plots and their applications". *The European Physical Journal Special Topics*, vol. 164, no. 1, pp. 13–22, 2008. doi: https://doi.org/10.1140/epjst/e2008-00830-8.

[13] Aloys Sipers, Paul J. A. Borm, and Ralf L. M. Peeters. "On the unique reconstruction of a signal from its unthresholded recurrence plot". *Physics Letters A*, vol. 375, no. 24, pp. 2309–2321, 2011. doi: https://doi.org/10.1016/j.physleta.2011.04.040.

[14] Aloys Sipers, Paul J. A. Borm, and Ralf L. M. Peeters. "Robust reconstruction of a signal from its unthresholded recurrence plot subject to disturbances". *Physics Letters A*, vol. 381, no. 6, pp. 604–615, 2017. doi:

https://doi.org/10.1016/j.physleta.2016.12.028.

[15] Warren S. Torgerson. "Multidimensional scaling: I. Theory and method". *Psychometrika*, vol. 17, no. 4, pp. 401–419, 1952. doi: https://doi.org/10.1007/BF02288916.

[16] Jonathan E. Atkins, Erik G. Boman, and Bruce Hendrickson. "A spectral algorithm for seriation and the consecutive ones problem". *SIAM Journal on Computing*, vol. 28, no. 1, pp. 297–310, 1998. doi: https://doi.org/10.1137/S0097539795285771.

[17] Gary M. Weiss. "WISDM Smartphone and Smartwatch Activity and Biometrics Dataset". *UCI Machine Learning Repository*, 2019. doi: https://doi.org/10.24432/C5HK59.

[18] Peter D. Welch. "The use of fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms". *IEEE Transactions on Audio and Electroacoustics*, vol. 15, no. 2, pp. 70–73, 1967. doi: https://doi.org/10.1109/TAU.1967.1161901.

[19] Frank Wilcoxon. "Individual comparisons by ranking methods". *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945. doi: https://doi.org/10.2307/3001968.

[20] Quinn McNemar. "Note on the sampling error of the difference between correlated proportions or percentages". *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947. doi: https://doi.org/10.1007/BF02295996.

[21] Anthony C. Davison and David V. Hinkley. *Bootstrap Methods and Their Application*. Cambridge University Press, 1997.

[22] Peter Hall. "On the bootstrap and likelihood-based confidence regions". *Biometrika*, vol. 74, no. 3, pp. 481–493, 1987. doi: https://doi.org/10.1093/biomet/74.3.481.

[23] Daniël Lakens. "Equivalence tests: A practical primer for t tests, correlations,

and meta-analyses". *Social Psychological and Personality Science*, vol. 8, no. 4, pp. 355–362, 2017. doi: https://doi.org/10.1177/1948550617697177.

[24] Donald J. Schuirmann. "A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability". *Journal of Pharmacokinetics and Biopharmaceutics*, vol. 15, no. 6, pp. 657–680, 1987. doi: https://doi.org/10.1007/BF01068419.

[25] Sture Holm. "A simple sequentially rejective multiple test procedure". *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.